2023-2 기계공학실험(1)

Mechanics of Materials module#1 manual

Tension test

■ Objectives

- How stress can be related to strain is shown by using experimental methods to determine the stress–strain diagram for a specific material.
- The behavior described by this diagram will then be discussed for materials that are commonly used in engineering.
- Mechanical properties and other tests that are related to the development of mechanics of materials will be discussed.

Tension test

■ Experiment Procedure

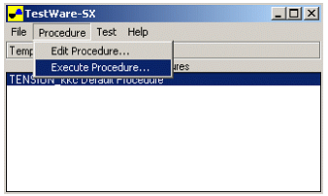
Check size of specimen → Check Fixture Alignment → Gripping (Crosshead) Specimen → Extensometer Installation



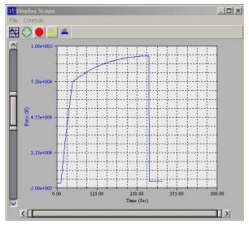


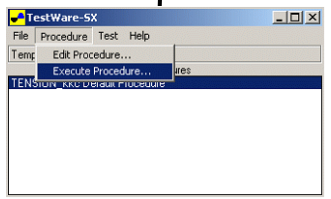

Gripping(Actuator) Specimen ← Force & Strain Zero Control ← Test Run(Control PC) ← Check Data and fracture






Save Test Data → Remove Extensometer → Un-Gripping Specimen → Test Stop(Control PC)



Check Young's Modulus E → Data Convert(σ-ε Curve)

Tension test

■ Cautions and Safety Guidelines

- Keep hands, head, and all other body parts away from the testing space when the crosshead is in motion
- Moving parts can produce dangerous forces corresponding to the system capacity. All operators should read the instruction manual thoroughly to ensure they operate the system correctly. Control access to the system to ensure only trained operators operate the system
- Due to risk of electric shock, never touch inside the controller unit, located in the testing machine
- Do not open the ball screw shaft protective cover unless lubricating the screw shaft. Do not leave the shaft cover open while performing tests. Keep hair, clothing, rags, and fingers from getting caught in the ball screw when lubricating the ball screw
- Keep the face and body away from the test specimen during tests. When the specimen breaks, flying debris could injure eyes or other body parts. Always use a safety guard or other measure to provide protection from broken specimens
- Due to risk of injury, or fire, wiring, maintenance, and inspection work should be performed only by a person with specialized expertise.
- Do not operate the system in a manner that subjects the crosshead to loads when the load cell is removed and do not operate the system in a manner that does not transmit the load to the load cell. Doing so would cause overloading to be undetected, which could result in damage to the frame or jigs
- Never operate the crosshead without crosshead limit switches in place. Doing so could cause the crosshead to hit the jig and damage the frame, jig, or load cell
- Before setting the input voltage, be sure to turn OFF the power switch and disconnect the power supply cable